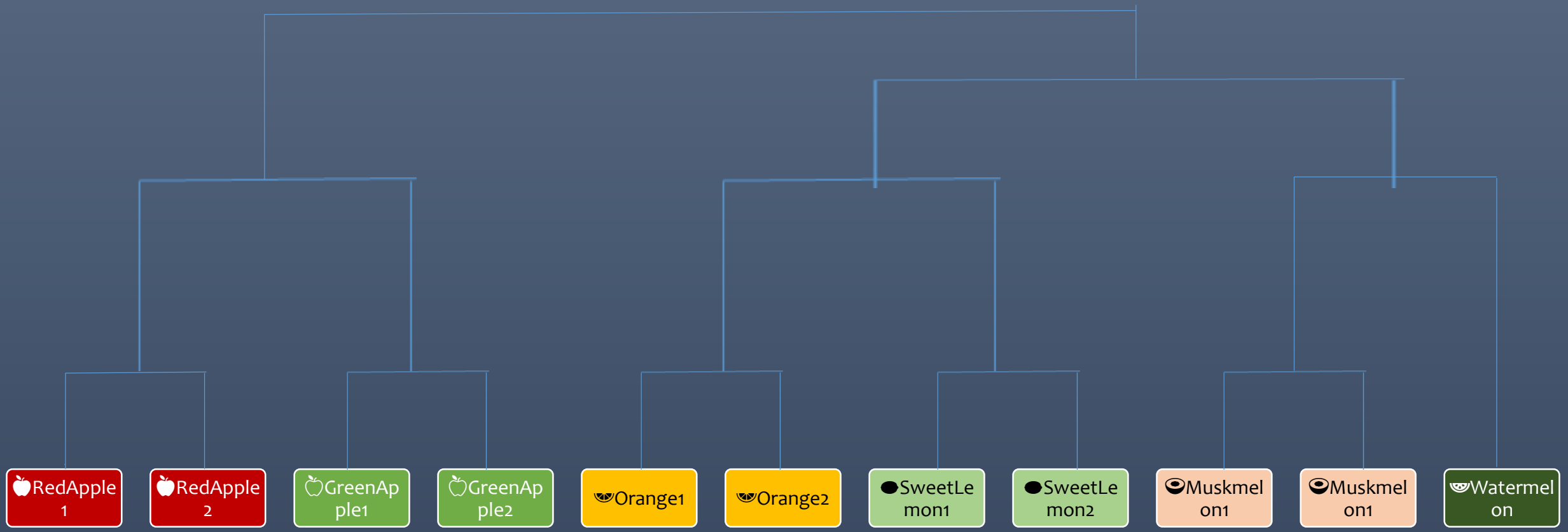
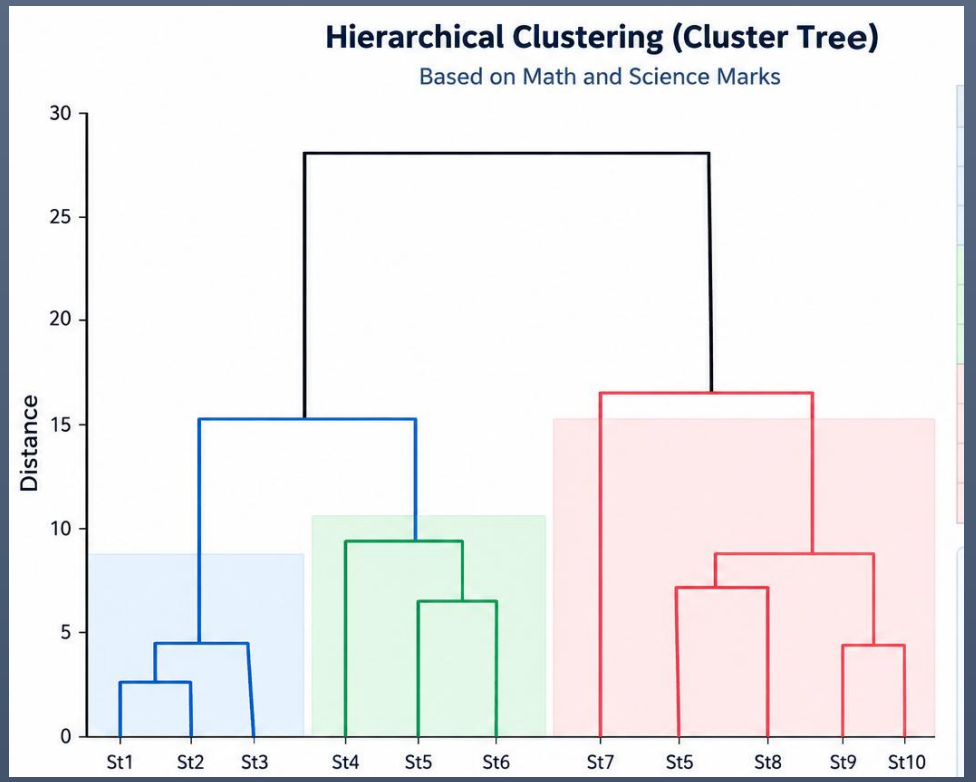
Fruit	Weight(g)	Sweetness
RedApple1	150	7
RedApple2	160	8
GreenApple	155	6
Orange1	120	6
Orange2	130	7
SweetLemon	125	5
Muskmelon	500	8
Watermelon	800	7


RedApple 1
RedApple 2
GreenApple1
GreenApple2
Orange1
Orange2
SweetLemon1
SweetLemon2
Muskmelon1
Muskmelon1
Watermelon

Student	Math	Science
St1	24	25
St2	29	20
St3	27	16
St4	69	68
St5	73	61
St6	62	70
St7	82	93
St8	83	84
St9	88	82
St10	99	84

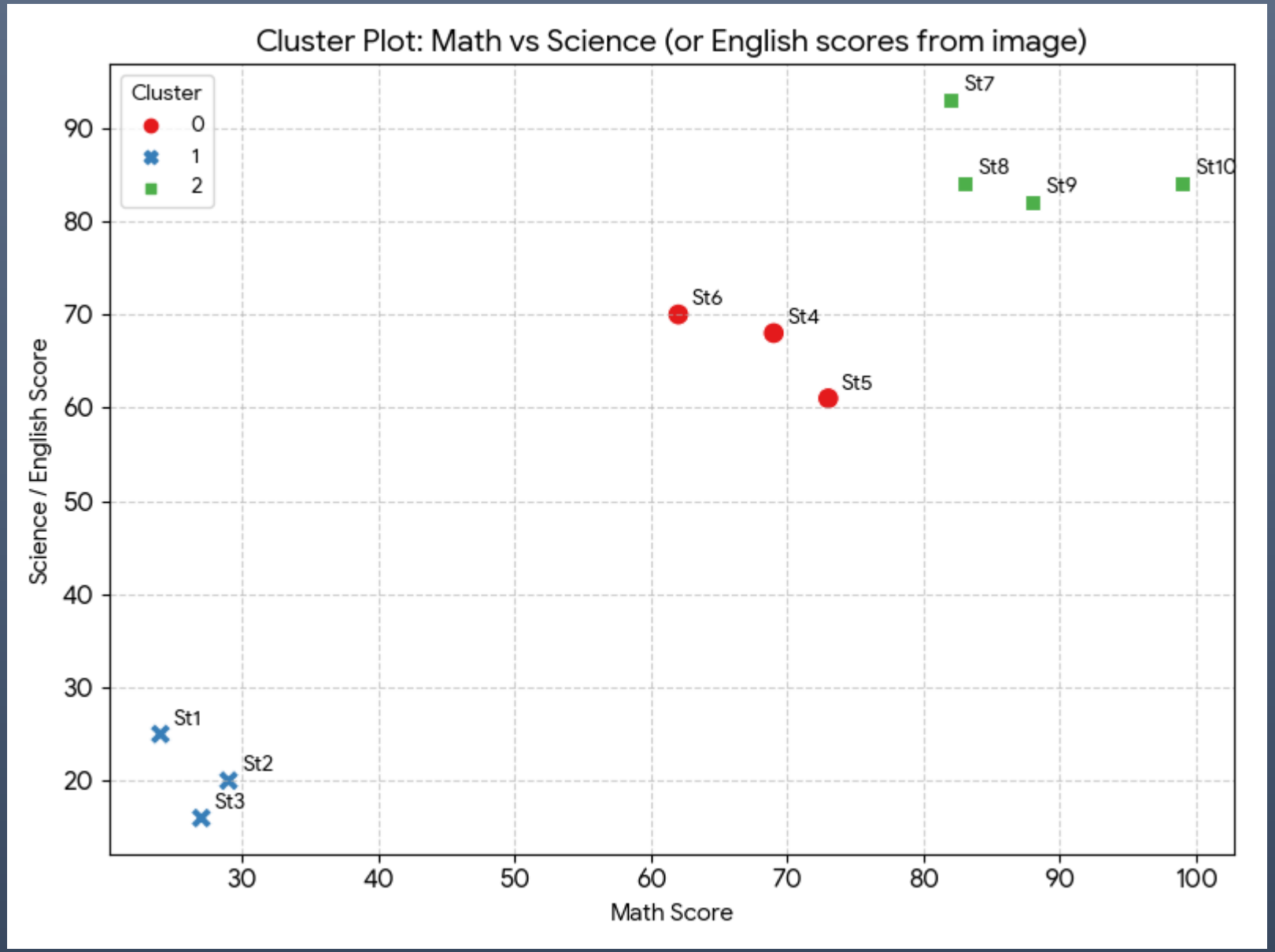


Student	Math	Science	Cluster
St1	24	25	1
St2	29	20	1
St3	27	16	1
St4	69	68	2
St5	73	61	2
St6	62	70	2
St7	82	93	3
St8	83	84	3
St9	88	82	3
St10	99	84	3

Cluster 1
(Low Marks)
St1, St2, St3

Cluster 2
(Medium Marks)
St4, St5, St6

Cluster 3
(High Marks)
St7, St8, St9, St10


Cluster Plot: Math vs Science (or English scores from image)
Cluster
0
1
2
St1
St2
St3
St4
St5
St6
St7
St8
St9
St10
Science / English Score
Math Score
20
30
40
50
60
70
80
90
30
40
50
60
70
80
90
100

The Euclidean distance between two points (x_1, y_1) and (x_2, y_2) is given by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Student	Math	Science
St1	24	25
St2	29	20
St3	27	16
St4	69	68
St5	73	61
St6	62	70
St7	82	93
St8	83	84
St9	88	82
St10	99	84

1. St-1 and St-2

$$d = \sqrt{(24 - 29)^2 + (25 - 20)^2}$$

$$= \sqrt{(-5)^2 + (5)^2}$$

$$= \sqrt{25 + 25}$$

$$= \sqrt{50} = 7.07$$

2. St-1 and St-3

$$d = \sqrt{(24 - 27)^2 + (25 - 16)^2}$$

$$= \sqrt{(-3)^2 + 9^2}$$

$$= \sqrt{9 + 81}$$

$$= \sqrt{90} = 9.49$$

3. St-1 and St-4

$$d = \sqrt{(24 - 69)^2 + (25 - 68)^2}$$

$$= \sqrt{(-45)^2 + (-43)^2}$$

$$= \sqrt{2025 + 1849}$$

$$= \sqrt{3874} = 62.24$$

4. St-1 and St-5

$$d = \sqrt{(24 - 73)^2 + (25 - 61)^2}$$

$$= \sqrt{(-49)^2 + (-36)^2}$$

$$= \sqrt{2401 + 1296}$$

$$= \sqrt{3697} = 60.80$$

5. St-1 and St-6

$$d = \sqrt{(24 - 62)^2 + (25 - 70)^2}$$

$$= \sqrt{(-38)^2 + (-45)^2}$$

$$= \sqrt{1444 + 2025}$$

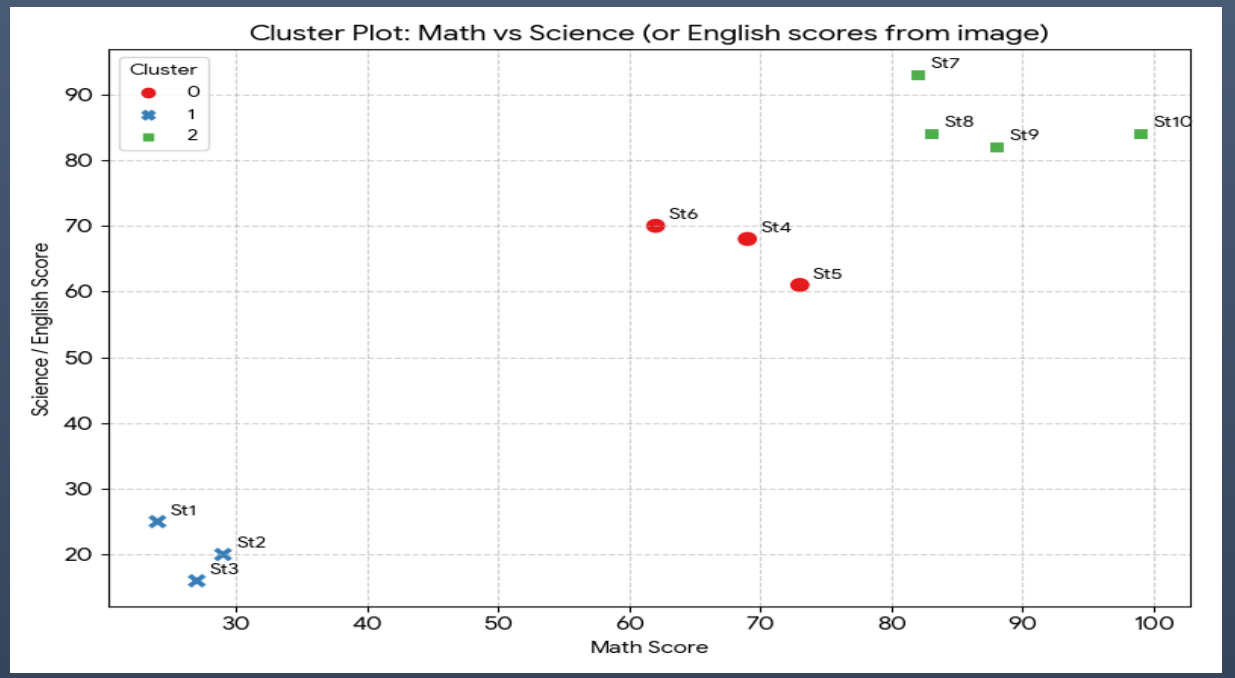
$$= \sqrt{3469} = 58.89$$

Student	To C_1(St-1)
St-1	-
St-2	7.1
St-3	9.5
St-4	62.2
St-5	60.8
St-6	58.9
St-7	89.4
St-8	83.4
St-9	85.7
St-10	95.4

Student	To C_2(St-10)
St-1	95.4
St-2	94.9
St-3	99.0
St-4	34.0
St-5	34.7
St-6	39.6
St-7	19.2
St-8	16.0
St-9	11.2
St-10	-

Smaller Distance
-
7.1
9.5
34.0
34.7
39.6
19.2
16.0
11.2
-

Distance^2
-
50.0
90.0
1,156.0
1,205.0
1,565.0
370.0
256.0
125.0
-

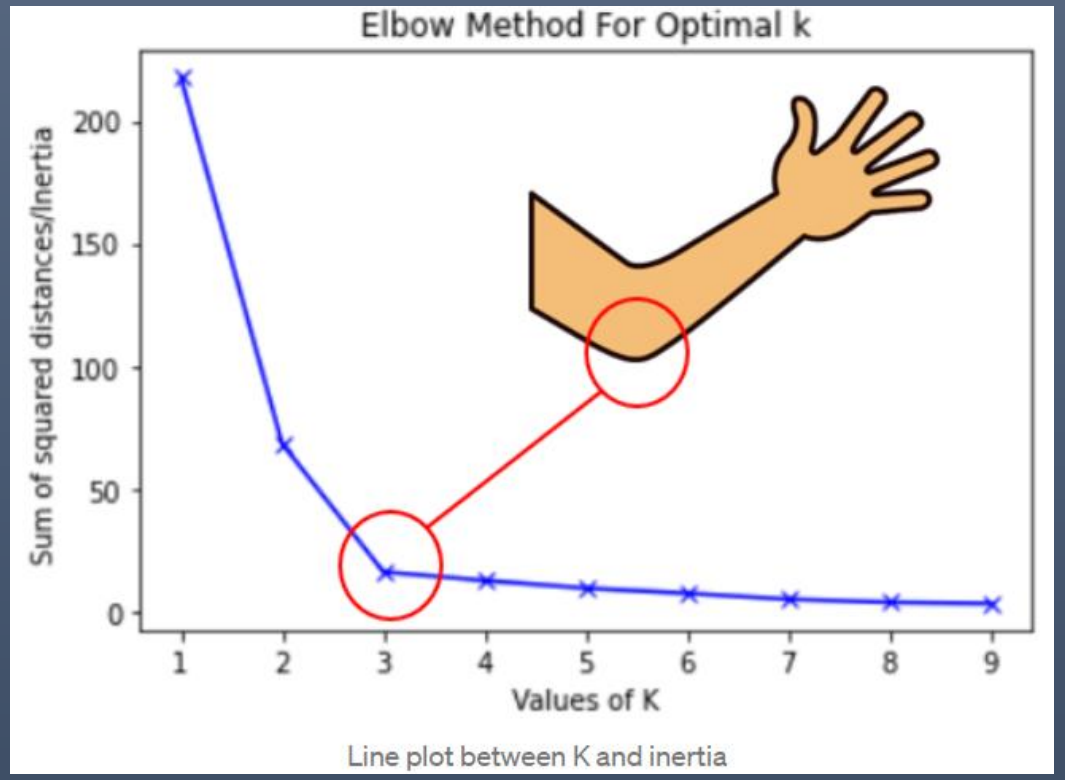


Sdnt	Math	Sce
St1	24	25
St2	29	20
St3	27	16
St4	69	68
St5	73	61
St6	62	70
St7	82	93
St8	83	84
St9	88	82
St10	99	84

Student	St-1	St-2	St-3	St-4	St-5	St-6	St-7	St-8	St-9	St-10
St-1	-	7.10	9.20	62.20	64.90	58.00	89.50	83.50	85.40	94.60
St-2	7.10	-	5.00	62.20	63.10	59.00	89.80	84.00	85.40	91.60
St-3	9.20	5.00	-	66.50	67.30	63.50	93.30	88.30	89.20	96.10
St-4	62.20	62.20	66.50	-	7.80	7.30	27.80	20.50	24.40	32.00
St-5	64.90	63.10	67.30	7.80	-	14.90	33.30	23.20	20.60	26.00
St-6	58.00	59.00	63.50	7.30	14.90	-	31.90	23.70	30.00	38.00
St-7	89.50	89.80	93.30	27.80	33.30	31.90	-	9.10	10.30	18.40
St-8	83.50	84.00	88.30	20.50	23.20	23.70	9.10	-	5.80	16.30
St-9	85.40	85.40	89.20	24.40	20.60	30.00	10.30	5.80	-	11.40
St-10	94.60	91.60	96.10	32.00	26.00	38.00	18.40	16.30	11.40	-

EL-Bow:

WCSS-Within cluster sum of squared distance



Student	St-1	St-2	St-3	St-4	St-5	St-6	St-7	St-8	St-9	St-10
St-1	-	7.10	9.20	62.20	64.90	58.00	89.50	83.50	85.40	94.60
St-2	7.10	-	5.00	62.20	63.10	59.00	89.80	84.00	85.40	91.60
St-3	9.20	5.00	-	66.50	67.30	63.50	93.30	88.30	89.20	96.10
St-4	62.20	62.20	66.50	-	7.80	7.30	27.80	20.50	24.40	32.00
St-5	64.90	63.10	67.30	7.80	-	14.90	33.30	23.20	20.60	26.00
St-6	58.00	59.00	63.50	7.30	14.90	-	31.90	23.70	30.00	38.00
St-7	89.50	89.80	93.30	27.80	33.30	31.90	-	9.10	10.30	18.40
St-8	83.50	84.00	88.30	20.50	23.20	23.70	9.10	-	5.80	16.30
St-9	85.40	85.40	89.20	24.40	20.60	30.00	10.30	5.80	-	11.40
St-10	94.60	91.60	96.10	32.00	26.00	38.00	18.40	16.30	11.40	-

Student	Math	Science
St1	24	25
St2	29	20
St3	27	16
St4	69	68
St5	73	61
St6	62	70
St7	82	93
St8	83	84
St9	88	82
St10	99	84

Student	St-1	St-2	St-3		St-4	St-5	St-6		St-7	St-8	St-9	St-10
St-1	-	7.1	9.2		62.2	64.9	58.0		89.5	83.5	85.4	94.6
St-2	7.1	-	5.0		62.2	63.1	59.0		89.8	84.0	85.4	91.6
St-3	9.2	5.0	-		66.5	67.3	63.5		93.3	88.3	89.2	96.1
St-4	62.2	62.2	66.5		-	7.8	7.3		27.8	20.5	24.4	32.0
St-5	64.9	63.1	67.3		7.8	-	14.9		33.3	23.2	20.6	26.0
St-6	58.0	59.0	63.5		7.3	14.9	-		31.9	23.7	30.0	38.0
St-7	89.5	89.8	93.3		27.8	33.3	31.9		-	9.1	10.3	18.4
St-8	83.5	84.0	88.3		20.5	23.2	23.7		9.1	-	5.8	16.3
St-9	85.4	85.4	89.2		24.4	20.6	30.0		10.3	5.8	-	11.4
St-10	94.6	91.6	96.1		32.0	26.0	38.0		18.4	16.3	11.4	-

Group1:St-1,St-2,St-3

Group2:St-4,St-5,St-6

Group3:St-7,St-8,St-9,St-10

Student	Math	Science	Cluster
St1	24	25	1
St2	29	20	1
St3	27	16	1
St4	69	68	2
St5	73	61	2
St6	62	70	2
St7	82	93	3
St8	83	84	3
St9	88	82	3
St10	99	84	3